

POLI666 Manual Replication - MASTER FILE

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Setup and data loading

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

set.seed(666)

library(tidyverse)
library(haven)
library(lmtest)
library(sandwich)
library(modelsummary)
library(kableExtra)
library(psych)
library(lavaan)

data <- read_dta("moscowpopsurvey.dta", encoding = "Windows-1251")
```

Cleaning data: Constructing variables

Building three time periods

```
data <- data %>%
  rename(day = DATA) %>%
  mutate(
    period = ifelse(day > 24 | day < 5, 1, 0),
    dec25 = ifelse(MONTH == 12 & day == 25, 1, 0),
    period1 = period - dec25,
    period2 = ifelse(day > 4 & day == 11, 1, 0),
    electday = ifelse(MONTH == 12 & day == 4, 1, 0),
    #electday = "Election Day"
    protestday = ifelse(MONTH == 12 & day == 10, 1, 0),
    #protestday = "Protest Day"
    period1min = period1 - electday,
    period2min = period2 - protestday,
    #period2min = "Post-Election"
    period3 = ifelse(day > 10 & day < 26 & MONTH == 12, 1, 0),
    #period3 = "Post-Protest"
```

Building trust measures

```

data <- data %>%
  mutate(trustarmy = case_match(q9a, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9a), #Army
    trustpolice = case_match(q9b, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9b), #Police
    trustfsb = case_match(q9c, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9c), #FSB
    trustcourt = case_match(q9d, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9d), #Court
    trustmungov = case_match(q9e, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9e), #City Gov't
    trustfedgov = case_match(q9f, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9f), #Fed Gov't
    trustduma = case_match(q9g, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9g), #Duma
    trustprocuracy = case_match(q9h, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9h), #Procuracy
    trustchurch = case_match(q9i, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9i), #Church
    #Church was not included in the original paper's analysis,
    #but is included here for subsequent analysis/checks.
    trustun = case_match(q9j, 99 ~ NA, 1 ~ 5, 2 ~ 4, 4 ~ 2, 5 ~ 1,
    .default = q9j)) #UN

#joining trust measures into gov trust index (trustgovind)

data <- data %>%
  mutate(trustgovind = (trustarmy + trustpolice + trustfsb + trustcourt +
    trustmungov + trustfedgov + trustduma +
    trustprocuracy)/8)
#Gov't Index (as built by Frye and Borisova 2019)

```

Mapping out/recoding potential confounders

```

data <- data %>%
  rename(age = q1,
    male = q2,
    wealth = q3,
    education = q5,
    permres = q59,
    nationality = q67) %>%
  mutate(male = ifelse(male == 2, 0, 1),
    wealth = case_match(wealth, 99 ~ NA, .default = wealth),
    education = case_match(education, 99 ~ NA, .default = education),
    permres = case_match(permres, 99 ~ NA, 2 ~ 0, 3 ~ 0,
    .default = permres),
    nationality = case_match(nationality, 2 ~ 0, 3 ~ 0, 4 ~ 0, 5 ~ 0,
    6 ~ 0, 7 ~ 0, 8 ~ 0, 9 ~ 0, 10 ~ 0, 11 ~ 0,
    99 ~ 0, .default = nationality),
    unemp = ifelse(q6 == 11 | q6 == 12 | q6 == 13, 1, 0),
    statesector = ifelse(q7 > 9 & q7 < 13, 1, 0),
    okrug = factor(okrug))

```

```
table(data$okrug)
```

```
##
##  1  2  3  4  5  6  7  8  9 10
## 107 146 173 200 173 232 202 168 120 29
```

Coding partisanship

```
#nesting each partisanship variable in its own mutate() function
#q62 = "Which party do you intend to vote for?"
#q64 = "Which party did you vote for in the previous election?"
data <- data %>%
  mutate(urplan = 0, urturn = 0,
         urplan = ifelse(!is.na(q62) & q62 == 6, 1, 0),
         urturn = ifelse(!is.na(q64) & q64 == 6, 1, 0),
         ur = urplan + urturn) %>% #united russia variable
  mutate(complan = 0, compturn = 0,
         complan = ifelse(!is.na(q62) & q62 == 3, 1, 0),
         compturn = ifelse(!is.na(q64) & q64 == 3, 1, 0),
         comp = complan + compturn) %>% #CPFR party (?)
  mutate(ldprplan = 0, ldprturn = 0,
         ldprplan = ifelse(!is.na(q62) & q62 == 4, 1, 0),
         ldprturn = ifelse(!is.na(q64) & q64 == 4, 1, 0),
         ldpr = ldprplan + ldprturn) %>% #LDPR party
  mutate(yabplan = 0, yabturn = 0,
         yabplan = ifelse(!is.na(q62) & q62 == 1, 1, 0),
         yabturn = ifelse(!is.na(q64) & q64 == 1, 1, 0), #Yabloko party
         yab = yabplan + yabturn) %>%
  mutate(patplan = 0, patturn = 0,
         patplan = ifelse(!is.na(q62) & q62 == 2, 1, 0),
         patturn = ifelse(!is.na(q64) & q64 == 2, 1, 0),
         pat = patplan + patturn) %>% #Unsure of party
  mutate(srplan = 0, srturn = 0,
         srplan = ifelse(!is.na(q62) & q62 == 5, 1, 0),
         srturn = ifelse(!is.na(q64) & q64 == 5, 1, 0),
         sr = srplan + srturn) %>% #Fair Russia party
  mutate(rcplan = 0, rcturn = 0,
         rcplan = ifelse(!is.na(q62) & q62 == 7, 1, 0),
         rcturn = ifelse(!is.na(q64) & q64 == 7, 1, 0),
         rc = rcplan + rcturn) %>% #Unsure of party
  mutate(dkplan = ifelse(!is.na(q62) & q62 == 99, 1, 0),
         dkturn = ifelse(!is.na(q64) & q64 == 99, 1, 0),
         dk = dkplan + dkturn) %>% #people who "don't know" (q62/q64 = 99)
  mutate(nopart = 0,
         nopart = ifelse((!is.na(q61) & q61 == 2) | (!is.na(q63) & q63 == 2) |
                        (!is.na(q62) & q62 == 0), 1, 0)) #no party

#q61 = "Do you intend to vote in the upcoming (Duma) elections?"
#q63 = "Did you vote in the previous (Duma) elections?"
```

Coding non-response variables

```
data <- data %>%
  mutate(trustgovind.nr = case_when(trustgovind >= 0.85 ~ 1, NA ~ 1, TRUE ~ 0),
         trustarmy.nr = case_match(q9a, 99 ~ 1, .default = 0),
         trustpolice.nr = case_match(q9b, 99 ~ 1, .default = 0),
         trustfsb.nr = case_match(q9c, 99 ~ 1, .default = 0),
         trustcourt.nr = case_match(q9d, 99 ~ 1, .default = 0),
         trustmungov.nr = case_match(q9e, 99 ~ 1, .default = 0),
         trustfedgov.nr = case_match(q9f, 99 ~ 1, .default = 0),
         trustduma.nr = case_match(q9g, 99 ~ 1, .default = 0),
         trustprocuracy.nr = case_match(q9h, 99 ~ 1, .default = 0),
         trustchurch.nr = case_match(q9i, 99 ~ 1, .default = 0),
         trustun.nr = case_match(q9j, 99 ~ 1, .default = 0))
```

Results

Table 1

```
#period 1 vs. period 2
p12.data <- data %>% filter(period3 != 1 & electday != 1 & protestday !=1)

#NOTE - can later pull:
#result$statistic - t-stat
#result$p.value - p-value
#result$conf.int ~ 95% CI
#result$estimate - mean(s) of the group(s)
#result$parameter - df
#result$stderr - SE

p12.army <- t.test(trustarmy ~ period1min, data = p12.data)
p12.police <- t.test(trustpolice ~ period1min, data = p12.data)
p12.fsb <- t.test(trustfsb ~ period1min, data = p12.data)
p12.court <- t.test(trustcourt ~ period1min, data = p12.data)
p12.mungov <- t.test(trustmungov ~ period1min, data = p12.data)
p12.fedgov <- t.test(trustfedgov ~ period1min, data = p12.data)
p12.duma <- t.test(trustduma ~ period1min, data = p12.data)
p12.procuracy <- t.test(trustprocuracy ~ period1min, data = p12.data)
p12.un <- t.test(trustun ~ period1min, data = p12.data)
p12.trustgovind <- t.test(trustgovind ~ period1min, data = p12.data)

#period 1 vs. period 3
p13.data <- data %>% filter(period2min != 1 & electday != 1 & protestday !=1)

p13.army <- t.test(trustarmy ~ period1min, data = p13.data)
p13.police <- t.test(trustpolice ~ period1min, data = p13.data)
p13.fsb <- t.test(trustfsb ~ period1min, data = p13.data)
p13.court <- t.test(trustcourt ~ period1min, data = p13.data)
p13.mungov <- t.test(trustmungov ~ period1min, data = p13.data)
p13.fedgov <- t.test(trustfedgov ~ period1min, data = p13.data)
p13.duma <- t.test(trustduma ~ period1min, data = p13.data)
p13.procuracy <- t.test(trustprocuracy ~ period1min, data = p13.data)
p13.un <- t.test(trustun ~ period1min, data = p13.data)
```

```

p13.trustgovind <- t.test(trustgovind ~ period1min, data = p13.data)

#period 2 vs. period 3
p23.data <- data %>% filter(period1min != 1 & electday != 1 & protestday !=1)

p23.army <- t.test(trustarmy ~ period2min, data = p23.data)
p23.police <- t.test(trustpolice ~ period2min, data = p23.data)
p23.fsb <- t.test(trustfsb ~ period2min, data = p23.data)
p23.court <- t.test(trustcourt ~ period2min, data = p23.data)
p23.mungov <- t.test(trustmungov ~ period2min, data = p23.data)
p23.fedgov <- t.test(trustfedgov ~ period2min, data = p23.data)
p23.duma <- t.test(trustduma ~ period2min, data = p23.data)
p23.procuracy <- t.test(trustprocuracy ~ period2min, data = p23.data)
p23.un <- t.test(trustun ~ period2min, data = p23.data)
p23.trustgovind <- t.test(trustgovind ~ period2min, data = p23.data)

```

Table 2

```

#running all regression separately

lm.govindex <- lm(trustgovind ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.army <- lm(trustarmy ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.police <- lm(trustpolice ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.fsb <- lm(trustfsb ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.court <- lm(trustcourt ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.mungov <- lm(trustmungov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.fedgov <- lm(trustfedgov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.duma <- lm(trustduma ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.procuracy <- lm(trustprocuracy ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.un <- lm(trustun ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

#reporting in one table
modelsummary(list("Gov Index" = lm.govindex, "Army" = lm.army, "Police" = lm.police,
  "FSB" = lm.fsb, "Court" = lm.court, "Municipal Gov" = lm.mungov,

```

```

    "Federal Gov" = lm.fedgov, "Duma" = lm.duma,
    "Procuracy" = lm.procuracy, "United Nations" = lm.un),
  title = "Determinants of trust in institutions",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
kable_styling(latex_options = c("scale_down", "hold_position"))

```

Table 1: Determinants of trust in institutions

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.13* (0.05)	0.05 (0.06)	0.14* (0.06)	0.17** (0.06)	0.07 (0.06)	0.19*** (0.06)	0.21*** (0.06)	0.08 (0.06)	0.05 (0.06)	0.12+ (0.07)
Post-election	0.11 (0.09)	0.19+ (0.10)	0.12 (0.11)	0.01 (0.10)	0.15 (0.11)	0.04 (0.11)	-0.02 (0.11)	0.08 (0.10)	0.18+ (0.11)	-0.11 (0.11)
Protest day	0.22+ (0.13)	0.53*** (0.14)	0.35* (0.14)	0.18 (0.16)	0.30* (0.15)	0.20 (0.16)	0.07 (0.16)	0.03 (0.15)	0.15 (0.15)	0.00 (0.16)
Election day	-0.05 (0.12)	-0.23 (0.16)	0.09 (0.15)	0.11 (0.17)	0.07 (0.15)	-0.19 (0.14)	-0.25+ (0.15)	-0.18 (0.13)	0.03 (0.14)	-0.43** (0.16)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.088	0.066	0.052	0.085	0.057	0.072	0.085	0.062	0.054	0.042

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Extending measures of trust

Including the Orthodox Church

```

data <- data %>%
  mutate(trustgovind.withchurch = (trustarmy + trustpolice + trustfsb + trustcourt +
    trustmungov + trustfedgov + trustduma + trustprocuracy +
    trustchurch)/9)

lm.indexwithchurch <- lm(trustgovind.withchurch ~ period3 + period2min + protestday +
  electday + wealth + education + okrug, data = data)

lm.church <- lm(trustchurch ~ period3 + period2min + protestday + electday + wealth +
  education + okrug, data = data)

#reporting in one table
modelsummary(list("Gov Index (with Church)" = lm.indexwithchurch,
  "Church" = lm.church),
  title = "Accounting for the role of Church",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
kable_styling(latex_options = c("scale_down", "hold_position"))

```

Table 2: Accounting for the role of Church

	Gov Index (with Church)	Church
Post-protest	0.13** (0.05)	0.10+ (0.06)
Post-election	0.08 (0.09)	0.06 (0.11)
Protest day	0.17 (0.12)	0.03 (0.16)
Election day	-0.07 (0.12)	-0.30 (0.20)
Num.Obs.	1192	1445
R2	0.087	0.042

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Leave-one-out indices

```
data <- data %>%
  mutate(trustgovind.noarmy = (trustpolice + trustfsb + trustcourt + trustmungov +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nopolice = (trustarmy + trustfsb + trustcourt + trustmungov +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nofsb = (trustpolice + trustarmy + trustcourt + trustmungov +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nocourt = (trustpolice + trustfsb + trustarmy + trustmungov +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nomungov = (trustpolice + trustfsb + trustcourt + trustarmy +
    trustfedgov + trustduma + trustprocuracy)/7,
    trustgovind.nofedgov = (trustpolice + trustfsb + trustcourt + trustmungov +
    trustarmy + trustduma + trustprocuracy)/7,
    trustgovind.noduma = (trustpolice + trustfsb + trustcourt + trustmungov +
    trustfedgov + trustarmy + trustprocuracy)/7,
    trustgovind.noproc = (trustpolice + trustfsb + trustcourt + trustmungov +
    trustfedgov + trustduma + trustarmy)/7)

lm.noarmy <- lm(trustgovind.noarmy ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nopolice <- lm(trustgovind.nopolice ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nofsb <- lm(trustgovind.nofsb ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nocourt <- lm(trustgovind.nocourt ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nomungov <- lm(trustgovind.nomungov ~ period3 + period2min + protestday + electday +
  wealth + education + okrug, data = data)

lm.nofedgov <- lm(trustgovind.nofedgov ~ period3 + period2min + protestday + electday +
```

```

        wealth + education + okrug, data = data)

lm.noduma <- lm(trustgovind.noduma ~ period3 + period2min + protestday + electday +
        wealth + education + okrug, data = data)

lm.noproc <- lm(trustgovind.noproc ~ period3 + period2min + protestday + electday +
        wealth + education + okrug, data = data)

#reporting in one table
modelsummary(list("No Army" = lm.noarmy, "No Police" = lm.nopolice,
        "No FSB" = lm.nofsb, "No Court" = lm.nocourt,
        "No Municipal Gov" = lm.nomungov, "No Federal Gov" = lm.nofedgov,
        "No Duma" = lm.noduma, "No Procuracy" = lm.noproc),
        title = "Robustness check on Gov't Trust: Leave-one-out indices",
        vcov = "HC2",
        coef_map = c("period3" = "Post-protest",
        "period2min" = "Post-election", "protestday" = "Protest day",
        "electday" = "Election day"),
        stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
        output = "kableExtra") %>%
        kable_styling(latex_options = c("scale_down", "hold_position"))

```

Table 3: Robustness check on Gov't Trust: Leave-one-out indices

	No Army	No Police	No FSB	No Court	No Municipal Gov	No Federal Gov	No Duma	No Procuracy
Post-protest	0.14** (0.05)	0.13** (0.05)	0.11* (0.05)	0.13** (0.05)	0.12* (0.05)	0.12* (0.05)	0.12* (0.05)	0.14** (0.05)
Post-election	0.08 (0.10)	0.09 (0.09)	0.12 (0.08)	0.10 (0.09)	0.13 (0.09)	0.14 (0.09)	0.13 (0.09)	0.08 (0.09)
Protest day	0.16 (0.14)	0.18 (0.13)	0.22+ (0.12)	0.20 (0.13)	0.24+ (0.13)	0.26* (0.13)	0.27* (0.13)	0.21 (0.13)
Election day	-0.03 (0.12)	-0.07 (0.12)	-0.09 (0.12)	-0.07 (0.12)	-0.03 (0.12)	-0.03 (0.12)	-0.03 (0.12)	-0.05 (0.12)
Num.Obs.	1242	1235	1332	1243	1235	1236	1242	1260
R2	0.082	0.088	0.078	0.089	0.090	0.089	0.088	0.087

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Sub-indices (based on Schneider 2017)

```

#model 1: fedgov + duma + mungov
data <- data %>%
        mutate(alt.trust.index1 = (trustfedgov + trustduma + trustmungov)/3)

lm.alt1 <- lm(alt.trust.index1 ~ period3 + period2min + protestday + electday +
        wealth + education + okrug, data = data)

##NOTE: Schneider distinguishes between police and armed forces, but the role of
#the FSB seems ambiguous within that. for that reason, models 2-5 differentially
#include/exclude the FSB to see if that has any effect on results.

#model 2: fedgov + duma + police + army
data <- data %>%
        mutate(alt.trust.index2 = (trustfedgov + trustduma + trustpolice + trustarmy)/4)

```



```

lm.alt2 <- lm(alt.trust.index2 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#model 3: fedgov + дума + police + army + fsb
data <- data %>%
  mutate(alt.trust.index3 = (trustfedgov + trustduma + trustpolice + trustfsb +
                             trustarmy)/5)

lm.alt3 <- lm(alt.trust.index3 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#model 4: fedgov + дума + court + police
data <- data %>%
  mutate(alt.trust.index4 = (trustfedgov + trustduma + trustpolice + trustcourt)/4)

lm.alt4 <- lm(alt.trust.index4 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#model 5: fedgov + дума + court + police + fsb
data <- data %>%
  mutate(alt.trust.index5 = (trustfedgov + trustduma + trustpolice + trustcourt +
                             trustfsb)/4)

lm.alt5 <- lm(alt.trust.index5 ~ period3 + period2min + protestday + electday +
              wealth + education + okrug, data = data)

#reporting in one table
modelsummary(list("Model 1" = lm.alt1, "Model 2.1" = lm.alt2,
                  "Model 2.2" = lm.alt3,
                  "Model 3.1" = lm.alt4,
                  "Model 3.2" = lm.alt5),
              title = "Robustness check on Gov't Trust: Alternative sub-indices",
              caption = "Sub-indices constructed following Schneider (2017)",
              vcov = "HC2",
              coef_map = c("period3" = "Post-protest",
                           "period2min" = "Post-election", "protestday" = "Protest day",
                           "electday" = "Election day"),
              stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
              add_rows = tribble(
                ~term, ~"Model 1", ~"Model 2.1", ~"Model 2.2", ~"Model 3.1", ~"Model 3.2",
                "Fed Gov", "Yes", "Yes", "Yes", "Yes", "Yes",
                "Duma", "Yes", "Yes", "Yes", "Yes", "Yes",
                "Mun Gov", "Yes", " ", " ", " ", " ",
                "Police", " ", "Yes", "Yes", "Yes", "Yes",
                "Army", " ", "Yes", "Yes", " ", " ",
                "Court", " ", " ", " ", " ", "Yes", "Yes",
                "FSB", " ", " ", " ", "Yes", " ", "Yes"),
              output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))

```

Table 4: Sub-indices constructed following Schneider (2017)

	Model 1	Model 2.1	Model 2.2	Model 3.1	Model 3.2
Post-protest	0.16** (0.05)	0.12* (0.05)	0.14** (0.05)	0.13** (0.05)	0.18** (0.06)
Post-election	0.04 (0.10)	0.11 (0.08)	0.08 (0.09)	0.09 (0.09)	0.08 (0.12)
Protest day	0.12 (0.15)	0.26* (0.12)	0.23+ (0.13)	0.16 (0.13)	0.20 (0.17)
Election day	-0.21 (0.13)	-0.14 (0.12)	-0.06 (0.13)	-0.06 (0.12)	-0.01 (0.15)
Num.Obs.	1438	1417	1282	1388	1275
R2	0.079	0.085	0.094	0.070	0.082
Fed Gov	Yes	Yes	Yes	Yes	Yes
Duma	Yes	Yes	Yes	Yes	Yes
Mun Gov	Yes				
Police		Yes	Yes	Yes	Yes
Army		Yes	Yes		
Court				Yes	Yes
FSB			Yes		Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standardized index

```
data <- data %>%
  mutate(trustarmy.z = as.numeric(scale(trustarmy)),
         trustpolice.z = as.numeric(scale(trustpolice)),
         trustfsb.z = as.numeric(scale(trustfsb)),
         trustcourt.z = as.numeric(scale(trustcourt)),
         trustmungov.z = as.numeric(scale(trustmungov)),
         trustfedgov.z = as.numeric(scale(trustfedgov)),
         trustduma.z = as.numeric(scale(trustduma)),
         trustprocuracy.z = as.numeric(scale(trustprocuracy)),
         trustgovindex.standardized = (trustarmy.z + trustpolice.z + trustfsb.z +
                                       trustcourt.z + trustmungov.z + trustfedgov.z +
                                       trustduma.z + trustprocuracy.z)/8)

lm.standardized <- lm(trustgovindex.standardized ~ period3 + period2min + protestday +
                      electday + wealth + education + okrug, data = data)

modelsummary(lm.standardized,
  title = "Robustness check on Gov't Trust: Index with Z-Scored Items",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
               "period2min" = "Post-election", "protestday" = "Protest day",
               "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))
```

Table 5: Robustness check on Gov't Trust: Index with Z-Scored Items

	(1)
Post-protest	0.13* (0.05)
Post-election	0.11 (0.09)
Protest day	0.22+ (0.13)
Election day	-0.05 (0.12)
Num.Obs.	1232
R2	0.087

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Factor-score index

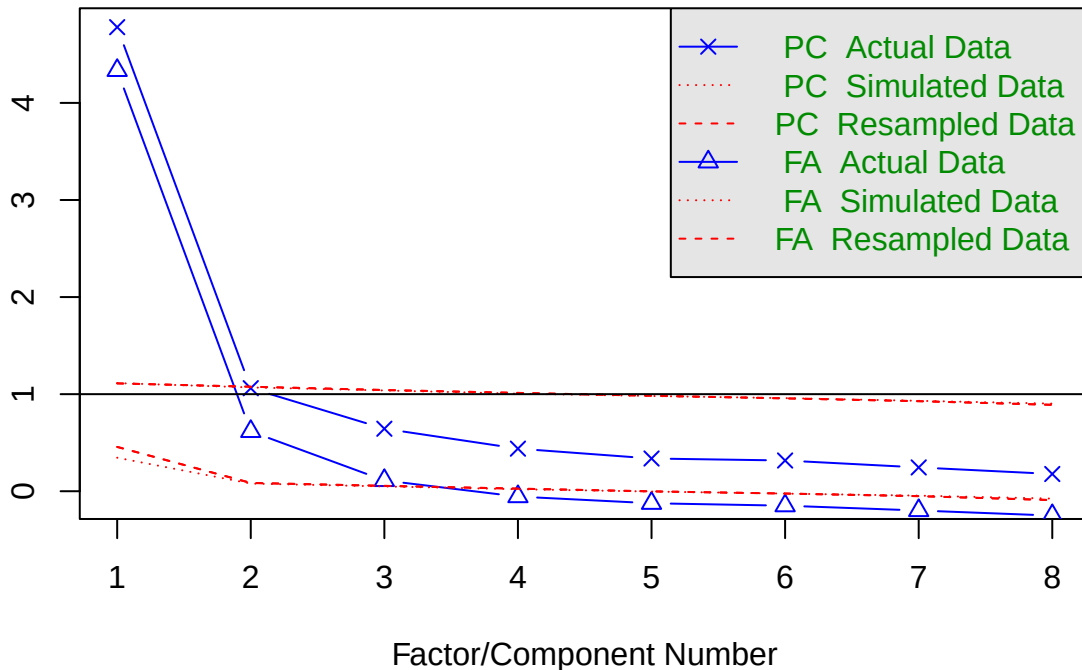
```
#Factor loading analysis
fa.analysis.data <- data %>%
  select(trustarmy, trustpolice, trustfsb, trustcourt, trustmungov,
         trustfedgov, trustduma, trustprocuracy)

fa.fit <- fa(fa.analysis.data, nfactors = 1, rotate = "oblimin", fm = "ml")

fa.parallel(fa.analysis.data, fm = "ml")
```

eigenvalues of principal components and factor analysis

Parallel Analysis Scree Plots



Paral-

lel analysis suggests that the number of factors = 3 and the number of components = 1

```
cr.alpha.data <- alpha(fa.analysis.data)
#crohnback's alpha ~ 0.903
data.frame(raw_alpha = cr.alpha.data$total$raw_alpha,
           std.alpha = cr.alpha.data$total$std.alpha)
```

```
raw_alpha std.alpha 1 0.903453 0.9027666
```

```
#Factor-score-weighted index
fa.fit$loadings
```

Loadings: ML1

```
trustarmy 0.530 trustpolice 0.691 trustfsb 0.703 trustcourt 0.680 trustmungov 0.820 trustfedgov 0.846
trustduma 0.817 trustprocuracy 0.718
```

ML1

SS loadings 4.284 Proportion Var 0.536

```
weights <- as.numeric(fa.fit$loadings)
weights.norm <- weights/sum(weights)
weights.norm
```

```
[1] 0.09125392 0.11901039 0.12114956 0.11710733 0.14129712 0.14580924 0.14074087 [8] 0.12363157
```

```
data <- data %>%
```

```
  mutate(trustgovind.factor.weight = (0.091*trustarmy) + (0.119*trustpolice) +
        (0.121*trustfsb) + (0.117*trustcourt) + (0.141*trustmungov) +
        (0.146*trustfedgov) + (0.141*trustduma) + (0.124*trustprocuracy))
```

```
#all weights add up to 1
```

```
lm.factor.weighted <- lm(trustgovind.factor.weight ~ period3 + period2min + protestday +
                        electday + wealth + education + okrug, data = data)
```

```
modelsummary(lm.factor.weighted,
  title = "Robustness check on Gov't Trust: Factor Loading Score-Weighted Index",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day"),
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))
```

```
#additional confirmatory factor analysis
```

```
model <- "trust =~ trustarmy + trustpolice + trustfsb + trustcourt +
        trustmungov + trustfedgov + trustduma + trustprocuracy"
```

```
model.fit <- cfa(model, data = data)
summary(model.fit, fit.measures = TRUE)
```

```
## lavaan 0.6-21 ended normally after 25 iterations
```

```
##
```

```
## Estimator ML
```

```
## Optimization method NLMINB
```

```
## Number of model parameters 16
```

```
##
```

Table 6: Robustness check on Gov't Trust: Factor Loading Score-Weighted Index

	(1)
Post-protest	0.13** (0.05)
Post-election	0.09 (0.09)
Protest day	0.19 (0.13)
Election day	-0.05 (0.12)
Num.Obs.	1232
R2	0.086

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
##                               Used      Total
##   Number of observations      1249      1550
##
## Model Test User Model:
##
##   Test statistic      1194.227
##   Degrees of freedom      20
##   P-value (Chi-square)      0.000
##
## Model Test Baseline Model:
##
##   Test statistic      6609.970
##   Degrees of freedom      28
##   P-value      0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)      0.822
##   Tucker-Lewis Index (TLI)      0.750
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)      -11465.307
##   Loglikelihood unrestricted model (H1)      NA
##
##   Akaike (AIC)      22962.614
##   Bayesian (BIC)      23044.696
##   Sample-size adjusted Bayesian (SABIC)      22993.873
##
## Root Mean Square Error of Approximation:
##
##   RMSEA      0.217
##   90 Percent confidence interval - lower      0.206
##   90 Percent confidence interval - upper      0.227
##   P-value H_0: RMSEA <= 0.050      0.000
##   P-value H_0: RMSEA >= 0.080      1.000
```

```
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                0.090
##
## Parameter Estimates:
##
##   Standard errors                Standard
##   Information                    Expected
##   Information saturated (h1) model Structured
##
## Latent Variables:
##
##           Estimate  Std.Err  z-value  P(>|z|)
## trust =~
##   trustarmy        1.000
##   trustpolice       1.235    0.070   17.707    0.000
##   trustfsb          1.278    0.073   17.624    0.000
##   trustcourt        1.239    0.070   17.722    0.000
##   trustmungov       1.513    0.077   19.563    0.000
##   trustfedgov       1.624    0.082   19.869    0.000
##   trustduma         1.552    0.079   19.555    0.000
##   trustprocuracy    1.299    0.072   18.145    0.000
##
## Variances:
##
##           Estimate  Std.Err  z-value  P(>|z|)
##   .trustarmy        0.772    0.032   24.198    0.000
##   .trustpolice       0.494    0.021   23.092    0.000
##   .trustfsb          0.546    0.024   23.152    0.000
##   .trustcourt        0.494    0.021   23.081    0.000
##   .trustmungov       0.281    0.014   19.913    0.000
##   .trustfedgov       0.248    0.014   18.360    0.000
##   .trustduma         0.297    0.015   19.946    0.000
##   .trustprocuracy    0.457    0.020   22.718    0.000
##   trust              0.300    0.031    9.732    0.000
```

##Comparative fit measurements - closer to 1 the better

#CFI: 0.822

#TLI: 0.75

#RMSEA: 0.217 (penalized model complexity)

#SRMR: 0.09 (difference between observed and model-implied correlations)

#NOTE: Borderline/poorer statistics more common with survey data (sensitive to N).

Mechanisms

Figure 1

```
data <- data %>%
  mutate(period123 = case_when(period1min == 1 ~ 1,
                                period2min == 1 ~ 2,
                                period3 == 1 ~ 3,
                                TRUE ~ NA_real_),
    opposition = yab + pat + sr + rc,
```

```

party = case_when(ur == 1 ~ 1,
                  comp == 1 ~ 2,
                  ldpr == 1 ~ 3,
                  opposition == 1 ~ 4,
                  nopart == 1 ~ 5,
                  dk == 1 ~ 6,
                  TRUE ~ 0),
partynames = case_when(party == 1 ~ "UR",
                       party == 2 ~ "CPRF",
                       party == 3 ~ "LDPR",
                       party == 4 ~ "Opposition",
                       party == 5 ~ "Non-voters",
                       party == 6 ~ "Undecided",
                       TRUE ~ NA_character_),
partynames = factor(partynames)) %>%
mutate(trustgovind1 = ifelse(period1min == 1, 1, 0),
       trustgovind2 = ifelse(period2min == 1, 1, 0),
       trustgovind3 = ifelse(period3 == 1, 1, 0))

data %>%
  group_by(partynames, period123) %>%
  summarize(mean.trust = mean(trustgovind, na.rm = TRUE)) %>%
  ungroup() %>%
  ggplot(aes(x = partynames,
             y = mean.trust,
             fill = factor(period123))) +
  geom_col(position = "dodge") +
  labs(y = "Trust in Government Index", fill = "Period")

```

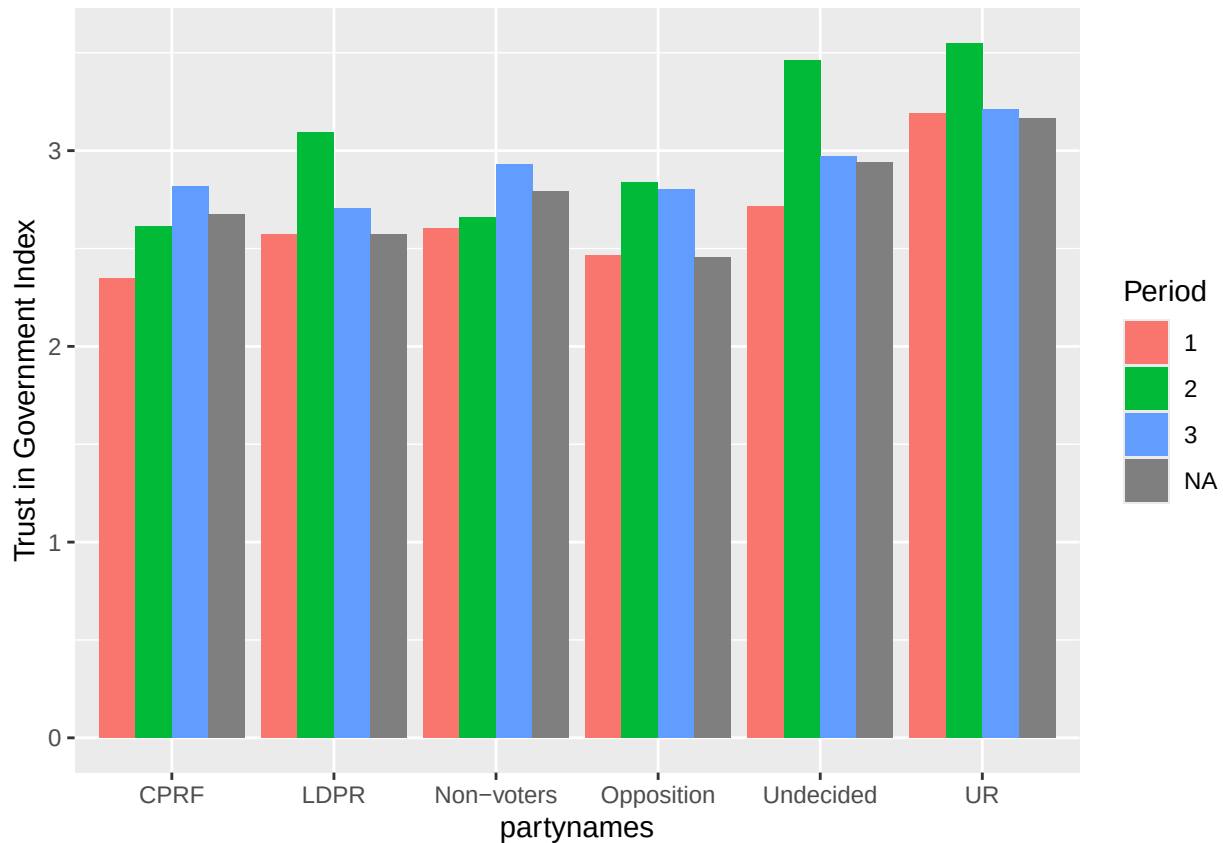


Table 3

```
data <- data %>%
  mutate(urafterprotest = ur*period3,
         urpostelection = ur*period2min,
         apolitical = nopart + dk,
         apolafterprotest = apolitical*period3,
         apolpostelection = apolitical*period2min)

lm.interact.govindex <- lm(trustgovind ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.army <- lm(trustarmy ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.police <- lm(trustpolice ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.fsb <- lm(trustfsb ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
```



```

apolitical + apolafterprotest + apolpostelection +
wealth + education + okrug, data = data)

lm.interact.court <- lm(trustcourt ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.mungov <- lm(trustmungov ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.fedgov <- lm(trustfedgov ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.duma <- lm(trustduma ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.procuracy <- lm(trustprocuracy ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

lm.interact.un <- lm(trustun ~ period3 + period2min + protestday + electday +
  ur + urafterprotest + urpostelection +
  apolitical + apolafterprotest + apolpostelection +
  wealth + education + okrug, data = data)

modelsummary(list("Gov Index" = lm.interact.govindex, "Army" = lm.interact.army, "Police" = lm.interact.m
  "FSB" = lm.interact.fsb, "Court" = lm.interact.court, "Municipal Gov" = lm.interact.m
  "Federal Gov" = lm.interact.fedgov, "Duma" = lm.interact.duma,
  "Procuracy" = lm.interact.procuracy, "United Nations" = lm.interact.un),
  title = "Determinants of trust in institutions",
  vcov = "HC2",
  coef_map = c("period3" = "Post-protest",
    "period2min" = "Post-election", "protestday" = "Protest day",
    "electday" = "Election day", "ur" = "United Russia", "urafterprotest" = "United Russia x P
  stars = TRUE, fmt = 2, gof_map = c("nobs", "r.squared"),
  output = "kableExtra") %>%
  kable_styling(latex_options = c("scale_down", "hold_position"))

```

Figure 2

```

coefs.robust <- coeftest(lm.interact.govindex, vcov = vcovHC(lm.interact.govindex, type = "HC2"))

vars <- c("period3", "period2min", "protestday", "electday", "ur", "urafterprotest", "urpostelection",
var.names <- c("period3" = "Post-Protest",
  "period2min" = "Post-Election",

```

Table 7: Determinants of trust in institutions

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.18* (0.08)	0.08 (0.09)	0.08 (0.09)	0.17+ (0.10)	0.11 (0.09)	0.28** (0.09)	0.26** (0.09)	0.21* (0.09)	0.13 (0.09)	0.08 (0.10)
Post-election	0.09 (0.13)	0.25* (0.13)	0.16 (0.14)	-0.01 (0.16)	0.16 (0.16)	-0.09 (0.15)	-0.13 (0.14)	-0.01 (0.14)	0.06 (0.16)	-0.11 (0.17)
Protest day	0.22+ (0.13)	0.52*** (0.14)	0.35* (0.14)	0.19 (0.16)	0.28+ (0.15)	0.22 (0.16)	0.09 (0.16)	0.03 (0.15)	0.14 (0.16)	0.02 (0.17)
Election day	-0.01 (0.11)	-0.20 (0.15)	0.13 (0.15)	0.15 (0.16)	0.10 (0.15)	-0.14 (0.15)	-0.20 (0.15)	-0.12 (0.14)	0.08 (0.14)	-0.43** (0.16)
United Russia	0.59*** (0.09)	0.42*** (0.13)	0.58*** (0.11)	0.58*** (0.12)	0.40*** (0.12)	0.66*** (0.11)	0.77*** (0.11)	0.74*** (0.12)	0.49*** (0.12)	0.13 (0.14)
United Russia x Post-Protest	-0.14 (0.14)	-0.07 (0.17)	-0.12 (0.16)	-0.14 (0.18)	-0.19 (0.17)	-0.17 (0.16)	-0.19 (0.16)	-0.34* (0.17)	-0.24 (0.18)	-0.07 (0.20)
United Russia x Post-Election	0.12 (0.20)	0.09 (0.21)	0.15 (0.23)	0.21 (0.25)	0.24 (0.21)	0.25 (0.21)	0.27 (0.21)	0.30 (0.22)	0.30 (0.24)	0.17 (0.26)
Apolitical	0.17* (0.07)	0.16+ (0.09)	0.09 (0.08)	0.12 (0.09)	0.16+ (0.09)	0.15+ (0.08)	0.15+ (0.08)	0.20* (0.08)	0.21* (0.08)	-0.08 (0.10)
Apolitical x Post-Protest	-0.05 (0.10)	-0.04 (0.12)	0.16 (0.11)	0.06 (0.12)	-0.02 (0.11)	-0.14 (0.11)	-0.06 (0.12)	-0.18 (0.11)	-0.10 (0.11)	0.12 (0.14)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.134	0.082	0.084	0.111	0.070	0.109	0.132	0.099	0.070	0.044

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```

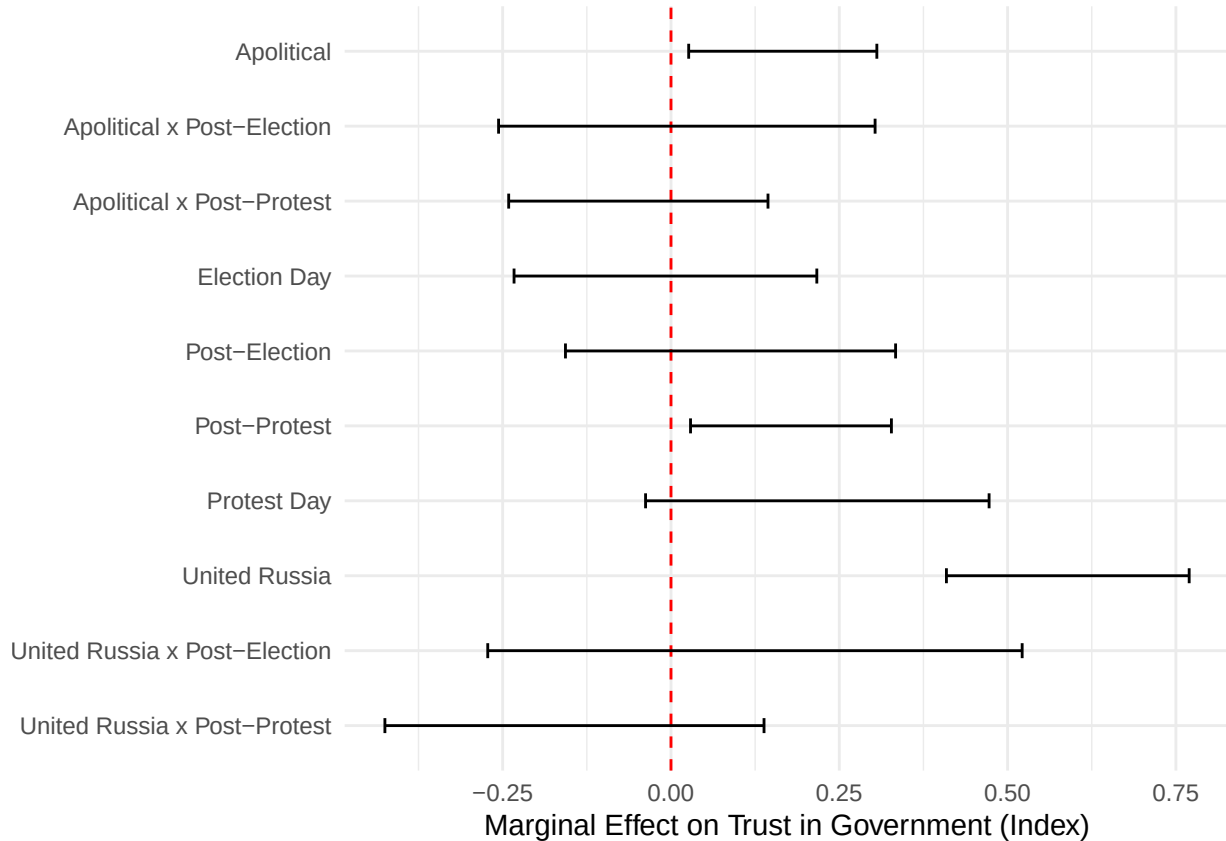
    "protestday" = "Protest Day",
    "electday" = "Election Day",
    "ur" = "United Russia",
    "urafterprotest" = "United Russia x Post-Protest",
    "urpostelection" = "United Russia x Post-Election",
    "apolitical" = "Apolitical",
    "apolafterprotest" = "Apolitical x Post-Protest",
    "apolpostelection" = "Apolitical x Post-Election")

margeffects.data <- data.frame(vars = vars,
                               beta = coefs.robust[vars, "Estimate"],
                               se = coefs.robust[vars, "Std. Error"])

margeffects.data <- margeffects.data %>%
  mutate(ci.lower = beta - 1.96 * se,
         ci.upper = beta + 1.96 * se,
         var.label = factor(var.names[vars]))

ggplot(margeffects.data, aes(x = beta, y = var.label)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  geom_errorbar(aes(xmin = ci.lower, xmax = ci.upper), height = 0.2) +
  scale_y_discrete(limits = rev) +
  labs(x = "Marginal Effect on Trust in Government (Index)", y = NULL) +
  theme_minimal()

```



Appendices

Appendix I: (Balance test:) Distribution of respondents across districts by periods

```
## Period 1 vs. Period 2 test without electday and protestday variables

balance.data1a <- data %>%
  filter(period3 != 1, electday != 1, protestday != 1)

t1.1a <- t.test(balance.data1a$okrug == 1 ~ balance.data1a$period1min, var.equal = TRUE)
t1.2a <- t.test(balance.data1a$okrug == 2 ~ balance.data1a$period1min, var.equal = TRUE)
t1.3a <- t.test(balance.data1a$okrug == 3 ~ balance.data1a$period1min, var.equal = TRUE)
t1.4a <- t.test(balance.data1a$okrug == 4 ~ balance.data1a$period1min, var.equal = TRUE)
t1.5a <- t.test(balance.data1a$okrug == 5 ~ balance.data1a$period1min, var.equal = TRUE)
t1.6a <- t.test(balance.data1a$okrug == 6 ~ balance.data1a$period1min, var.equal = TRUE)
t1.7a <- t.test(balance.data1a$okrug == 7 ~ balance.data1a$period1min, var.equal = TRUE)
t1.8a <- t.test(balance.data1a$okrug == 8 ~ balance.data1a$period1min, var.equal = TRUE)
t1.9a <- t.test(balance.data1a$okrug == 9 ~ balance.data1a$period1min, var.equal = TRUE)
t1.10a <- t.test(balance.data1a$okrug == 10 ~ balance.data1a$period1min, var.equal = TRUE)

tests.1a <- list(t1.1a, t1.2a, t1.3a, t1.4a, t1.5a, t1.6a, t1.7a, t1.8a, t1.9a, t1.10a)

balance.tab1a <- data.frame(district = paste0("okrug", 1:10),
  mean.period2 = sapply(tests.1a, function(x) x$estimate[1]),
  mean.period1 = sapply(tests.1a, function(x) x$estimate[2]),
  diff = sapply(tests.1a,
```

```

function(x) x$estimate[2] - x$estimate[1]),
t.stat = sapply(tests.1a, function(x) x$statistic),
p.val = sapply(tests.1a, function(x) x$p.value),
row.names = NULL)

```

```
balance.tab1a
```

```

##      district mean.period2 mean.period1      diff      t.stat      p.val
## 1   okrug1     0.043750    0.20242915  0.15867915 -6.1016994 1.949313e-09
## 2   okrug2     0.053125    0.17004049  0.11691549 -4.5965718 5.302740e-06
## 3   okrug3     0.243750    0.02024291 -0.22350709  7.8477225 2.134040e-14
## 4   okrug4     0.162500    0.04453441 -0.11796559  4.5026997 8.153835e-06
## 5   okrug5     0.171875    0.08097166 -0.09090334  3.1907468 1.497805e-03
## 6   okrug6     0.109375    0.28340081  0.17402581 -5.4154462 9.055387e-08
## 7   okrug7     0.131250    0.11336032 -0.01788968  0.6411814 5.216648e-01
## 8   okrug8     0.081250    0.05668016 -0.02456984  1.1321572 2.580486e-01
## 9   okrug9     0.000000    0.00000000  0.00000000      NaN      NaN
## 10  okrug10     0.003125    0.02834008  0.02521508 -2.5340265 1.154504e-02

```

```
## Period 1 vs. Period 3 test without electday and protestday variables
```

```
balance.data1b <- data %>%
```

```
  filter(period2min != 1, electday != 1, protestday != 1)
```

```
t1.1b <- t.test(balance.data1b$okrug == 1 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.2b <- t.test(balance.data1b$okrug == 2 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.3b <- t.test(balance.data1b$okrug == 3 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.4b <- t.test(balance.data1b$okrug == 4 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.5b <- t.test(balance.data1b$okrug == 5 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.6b <- t.test(balance.data1b$okrug == 6 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.7b <- t.test(balance.data1b$okrug == 7 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.8b <- t.test(balance.data1b$okrug == 8 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.9b <- t.test(balance.data1b$okrug == 9 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
t1.10b <- t.test(balance.data1b$okrug == 10 ~ balance.data1b$period1min, var.equal = TRUE)
```

```
tests.1b <- list(t1.1b, t1.2b, t1.3b, t1.4b, t1.5b, t1.6b, t1.7b, t1.8b, t1.9b, t1.10b)
```

```
balance.tab1b <- data.frame(district = paste0("okrug", 1:10),
```

```
  mean.period3 = sapply(tests.1b, function(x) x$estimate[1]),
```

```
  mean.period1 = sapply(tests.1b, function(x) x$estimate[2]),
```

```
  diff = sapply(tests.1b,
```

```
    function(x) x$estimate[2] - x$estimate[1]),
```

```
  t.stat = sapply(tests.1b, function(x) x$statistic),
```

```
  p.val = sapply(tests.1b, function(x) x$p.value),
```

```
  row.names = NULL)
```

```
balance.tab1b
```

```

##      district mean.period3 mean.period1      diff      t.stat      p.val
## 1   okrug1     0.041246563    0.20242915  0.16118259 -9.176063 1.646963e-19
## 2   okrug2     0.095325390    0.17004049  0.07471510 -3.413053 6.616129e-04
## 3   okrug3     0.139321723    0.02024291 -0.11907881  5.302071 1.338403e-07
## 4   okrug4     0.150320807    0.04453441 -0.10578639  4.482565 8.006433e-06
## 5   okrug5     0.107241063    0.08097166 -0.02626940  1.229499 2.191014e-01
## 6   okrug6     0.119156737    0.28340081  0.16424407 -6.639210 4.574138e-11

```

```
## 7   okrug7  0.150320807   0.11336032 -0.03696048  1.496313 1.348083e-01
## 8   okrug8  0.096241980   0.05668016 -0.03956182  1.973773 4.861329e-02
## 9   okrug9  0.091659028   0.00000000 -0.09165903  4.988693 6.878433e-07
## 10  okrug10 0.009165903    0.02834008  0.01917418 -2.433129 1.509944e-02
```

Period 2 vs. Period 3 test without electday and protestday variables

```
balance.data1c <- data %>%
  filter(period1min != 1, electday != 1, protestday != 1)

t1.1c <- t.test(balance.data1c$okrug == 1 ~ balance.data1c$period3, var.equal = TRUE)
t1.2c <- t.test(balance.data1c$okrug == 2 ~ balance.data1c$period3, var.equal = TRUE)
t1.3c <- t.test(balance.data1c$okrug == 3 ~ balance.data1c$period3, var.equal = TRUE)
t1.4c <- t.test(balance.data1c$okrug == 4 ~ balance.data1c$period3, var.equal = TRUE)
t1.5c <- t.test(balance.data1c$okrug == 5 ~ balance.data1c$period3, var.equal = TRUE)
t1.6c <- t.test(balance.data1c$okrug == 6 ~ balance.data1c$period3, var.equal = TRUE)
t1.7c <- t.test(balance.data1c$okrug == 7 ~ balance.data1c$period3, var.equal = TRUE)
t1.8c <- t.test(balance.data1c$okrug == 8 ~ balance.data1c$period3, var.equal = TRUE)
t1.9c <- t.test(balance.data1c$okrug == 9 ~ balance.data1c$period3, var.equal = TRUE)
t1.10c <- t.test(balance.data1c$okrug == 10 ~ balance.data1c$period3, var.equal = TRUE)

tests.1c <- list(t1.1c, t1.2c, t1.3c, t1.4c, t1.5c, t1.6c, t1.7c, t1.8c, t1.9c, t1.10c)

balance.tab1c <- data.frame(district = paste0("okrug", 1:10),
  mean.period2 = sapply(tests.1c, function(x) x$estimate[1]),
  mean.period3 = sapply(tests.1c, function(x) x$estimate[2]),
  diff = sapply(tests.1c,
    function(x) x$estimate[2] - x$estimate[1]),
  t.stat = sapply(tests.1c, function(x) x$statistic),
  p.val = sapply(tests.1c, function(x) x$p.value),
  row.names = NULL)

balance.tab1c
```

##	district	mean.period2	mean.period3	diff	t.stat	p.val
## 1	okrug1	0.043750	0.04038005	-0.003369952	0.2576517	7.967214e-01
## 2	okrug2	0.053125	0.10332542	0.050200416	-2.6838115	7.382428e-03
## 3	okrug3	0.243750	0.08907363	-0.154676366	7.1097580	2.027643e-12
## 4	okrug4	0.162500	0.14014252	-0.022357482	0.9628888	3.358040e-01
## 5	okrug5	0.171875	0.09619952	-0.075675475	3.6015272	3.297323e-04
## 6	okrug6	0.109375	0.13064133	0.021266330	-0.9794621	3.275560e-01
## 7	okrug7	0.131250	0.14489311	0.013643112	-0.5962835	5.511022e-01
## 8	okrug8	0.081250	0.11163895	0.030388955	-1.5208412	1.285722e-01
## 9	okrug9	0.000000	0.13064133	0.130641330	-6.9285399	7.034682e-12
## 10	okrug10	0.003125	0.01306413	0.009939133	-1.4972290	1.346057e-01

Appendix II: (Balance tests:) Pre-treatment covariates

Period 1 vs. Period 2 test without electday and protestday variables

```
t2a.age <- t.test(balance.data1a$age ~ balance.data1a$period1min, var.equal = TRUE)
t2a.male <- t.test(balance.data1a$male ~ balance.data1a$period1min, var.equal = TRUE)
t2a.educ <- t.test(balance.data1a$education ~ balance.data1a$period1min, var.equal = TRUE)
t2a.wealth <- t.test(balance.data1a$wealth ~ balance.data1a$period1min, var.equal = TRUE)
t2a.permres <- t.test(balance.data1a$permres ~ balance.data1a$period1min, var.equal = TRUE)
```

```

t2a.nationality <- t.test(balance.data1a$nationality ~ balance.data1a$period1min, var.equal = TRUE)
t2a.unemp <- t.test(balance.data1a$unemp ~ balance.data1a$period1min, var.equal = TRUE)
t2a.statesector <- t.test(balance.data1a$statesector ~ balance.data1a$period1min, var.equal = TRUE)
t2a.ur <- t.test(balance.data1a$ur ~ balance.data1a$period1min, var.equal = TRUE)
t2a.comp <- t.test(balance.data1a$comp ~ balance.data1a$period1min, var.equal = TRUE)
t2a.opposition <- t.test(balance.data1a$opposition ~ balance.data1a$period1min, var.equal = TRUE)

tests.2a <- list(age = t2a.age, male = t2a.male, education = t2a.educ, wealth = t2a.wealth, permres = t2a.permres,
nationality = t2a.nationality, unemp = t2a.unemp, statesector = t2a.statesector, ur = t2a.ur, comp = t2a.comp,
opposition = t2a.opposition)

balance.tab2a <- data.frame(var.name = names(tests.2a),
                           mean.period2 = sapply(tests.2a, function(x) x$estimate[1]),
                           mean.period1 = sapply(tests.2a, function(x) x$estimate[2]),
                           diff = sapply(tests.2a,
                                           function(x) x$estimate[2] - x$estimate[1]),
                           t.stat = sapply(tests.2a, function(x) x$statistic),
                           p.val = sapply(tests.2a, function(x) x$p.value),
                           row.names = NULL)

```

```
balance.tab2a
```

##	var.name	mean.period2	mean.period1	diff	t.stat	p.val
## 1	age	43.51250000	42.23886640	-1.27363360	0.9846781	0.32520368
## 2	male	0.45937500	0.43319838	-0.02617662	0.6206120	0.53510517
## 3	education	5.36875000	5.55060729	0.18185729	-1.2350745	0.21731611
## 4	wealth	3.79682540	3.72839506	-0.06843034	1.1607705	0.24623339
## 5	permres	0.95937500	0.93522267	-0.02415233	1.2940743	0.19616846
## 6	nationality	0.85937500	0.90688259	0.04750759	-1.7280273	0.08452973
## 7	unemployment	0.05312500	0.06882591	0.01570091	-0.7798356	0.43581417
## 8	statesector	0.06481481	0.02197802	-0.04283679	2.0544772	0.04058528
## 9	ur	0.13750000	0.17004049	0.03254049	-1.0702611	0.28495878
## 10	comp	0.11562500	0.15789474	0.04226974	-1.4650520	0.14346295
## 11	opposition	0.16875000	0.13765182	-0.03109818	1.0131358	0.31142894

```
## Period 1 vs. Period 3 test without electday and protestday variables
```

```

t2b.age <- t.test(balance.data1b$age ~ balance.data1b$period1min, var.equal = TRUE)
t2b.male <- t.test(balance.data1b$male ~ balance.data1b$period1min, var.equal = TRUE)
t2b.educ <- t.test(balance.data1b$education ~ balance.data1b$period1min, var.equal = TRUE)
t2b.wealth <- t.test(balance.data1b$wealth ~ balance.data1b$period1min, var.equal = TRUE)
t2b.permres <- t.test(balance.data1b$permres ~ balance.data1b$period1min, var.equal = TRUE)
t2b.nationality <- t.test(balance.data1b$nationality ~ balance.data1b$period1min, var.equal = TRUE)
t2b.unemp <- t.test(balance.data1b$unemp ~ balance.data1b$period1min, var.equal = TRUE)
t2b.statesector <- t.test(balance.data1b$statesector ~ balance.data1b$period1min, var.equal = TRUE)
t2b.ur <- t.test(balance.data1b$ur ~ balance.data1b$period1min, var.equal = TRUE)
t2b.comp <- t.test(balance.data1b$comp ~ balance.data1b$period1min, var.equal = TRUE)
t2b.opposition <- t.test(balance.data1b$opposition ~ balance.data1b$period1min, var.equal = TRUE)

tests.2b <- list(age = t2b.age, male = t2b.male, education = t2b.educ, wealth = t2b.wealth, permres = t2b.permres,
nationality = t2b.nationality, unemp = t2b.unemp, statesector = t2b.statesector, ur = t2b.ur, comp = t2b.comp,
opposition = t2b.opposition)

balance.tab2b <- data.frame(var.name = names(tests.2b),
                           mean.period3 = sapply(tests.2b, function(x) x$estimate[1]),
                           mean.period1 = sapply(tests.2b, function(x) x$estimate[2]),
                           diff = sapply(tests.2b,
                                           function(x) x$estimate[2] - x$estimate[1]),

```

```
t.stat = sapply(tests.2b, function(x) x$statistic),
p.val = sapply(tests.2b, function(x) x$p.value),
row.names = NULL)
```

```
balance.tab2b
```

##	var.name	mean.period3	mean.period1	diff	t.stat	p.val
## 1	age	43.34555454	42.23886640	-1.106688140	0.9998746	0.317552273
## 2	male	0.47754354	0.43319838	-0.044345157	1.2608387	0.207587120
## 3	education	5.17231897	5.55060729	0.378288314	-3.1386406	0.001734473
## 4	wealth	3.67350746	3.72839506	0.054887599	-1.0832295	0.278905430
## 5	permres	0.95783685	0.93522267	-0.022614175	1.5268706	0.127029882
## 6	nationality	0.89275894	0.90688259	0.014123654	-0.6544679	0.512923057
## 7	unemployment	0.07516040	0.06882591	-0.006334492	0.3431957	0.731505262
## 8	statesector	0.04576976	0.02197802	-0.023791742	1.4468920	0.148274973
## 9	ur	0.13748854	0.17004049	0.032551943	-1.3175893	0.187867066
## 10	comp	0.11915674	0.15789474	0.038738000	-1.6553720	0.098083924
## 11	opposition	0.15765353	0.13765182	-0.020001707	0.7860945	0.431951551

Period 2 vs. Period 3 test without electday and protestday variables

```
t2c.age <- t.test(balance.data1c$age ~ balance.data1c$period3, var.equal = TRUE)
t2c.male <- t.test(balance.data1c$male ~ balance.data1c$period3, var.equal = TRUE)
t2c.educ <- t.test(balance.data1c$education ~ balance.data1c$period3, var.equal = TRUE)
t2c.wealth <- t.test(balance.data1c$wealth ~ balance.data1c$period3, var.equal = TRUE)
t2c.permres <- t.test(balance.data1c$permres ~ balance.data1c$period3, var.equal = TRUE)
t2c.nationality <- t.test(balance.data1c$nationality ~ balance.data1c$period3, var.equal = TRUE)
t2c.unemp <- t.test(balance.data1c$unemp ~ balance.data1c$period3, var.equal = TRUE)
t2c.statesector <- t.test(balance.data1c$statesector ~ balance.data1c$period3, var.equal = TRUE)
t2c.ur <- t.test(balance.data1c$ur ~ balance.data1c$period3, var.equal = TRUE)
t2c.comp <- t.test(balance.data1c$comp ~ balance.data1c$period3, var.equal = TRUE)
t2c.opposition <- t.test(balance.data1c$opposition ~ balance.data1c$period3, var.equal = TRUE)
```

```
tests.2c <- list(age = t2c.age, male = t2c.male, education = t2c.educ, wealth = t2c.wealth, permres = t2c.permres, nationality = t2c.nationality, unemployment = t2c.unemp, statesector = t2c.statesector, ur = t2c.ur, comp = t2c.comp, opposition = t2c.opposition)
```

```
balance.tab2c <- data.frame(var.name = names(tests.2c),
  mean.period2 = sapply(tests.2c, function(x) x$estimate[1]),
  mean.period3 = sapply(tests.2c, function(x) x$estimate[2]),
  diff = sapply(tests.2c,
    function(x) x$estimate[2] - x$estimate[1]),
  t.stat = sapply(tests.2c, function(x) x$statistic),
  p.val = sapply(tests.2c, function(x) x$p.value),
  row.names = NULL)
```

```
balance.tab2c
```

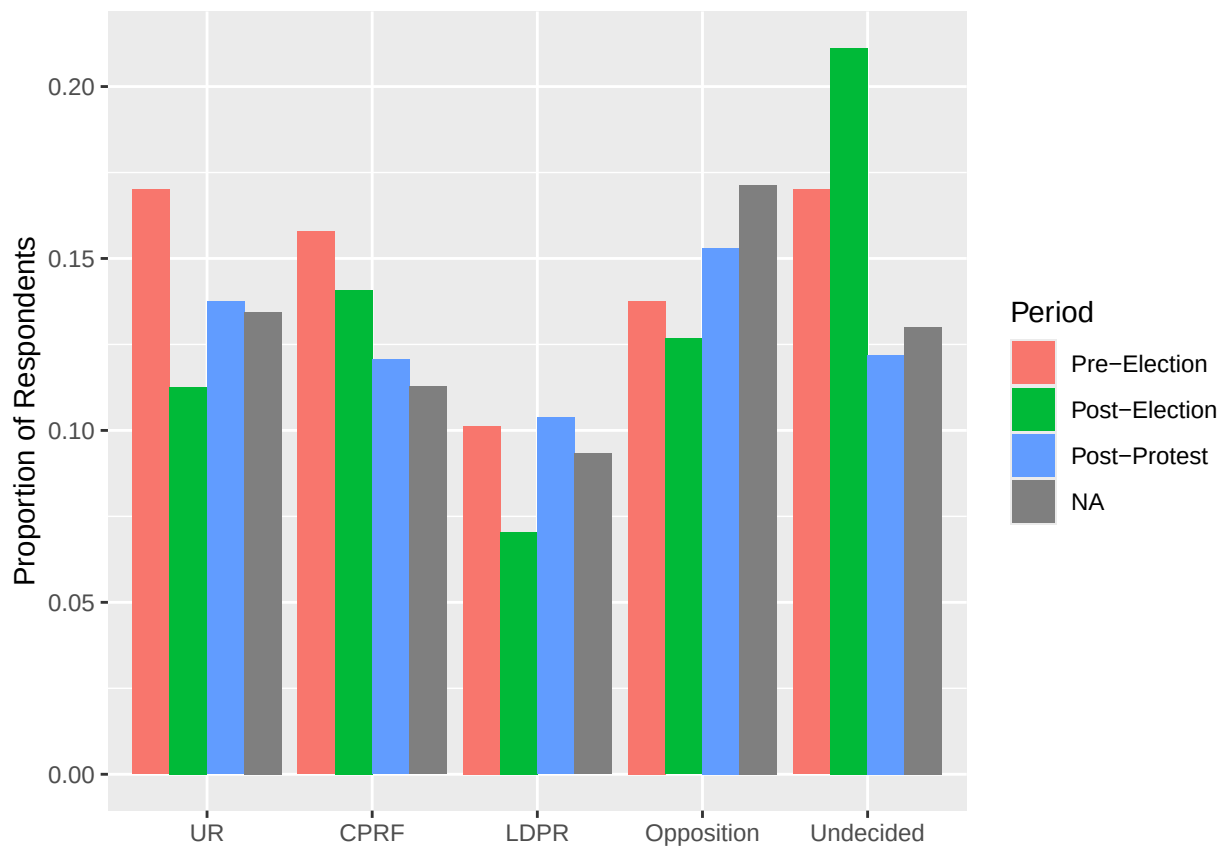
##	var.name	mean.period2	mean.period3	diff	t.stat	p.val
## 1	age	43.51250000	43.55700713	0.044507126	-0.04268694	0.9659584328
## 2	male	0.45937500	0.48218527	0.022810273	-0.69503987	0.4871695142
## 3	education	5.36875000	5.10688836	-0.261861639	2.35385979	0.0187456971
## 4	wealth	3.79682540	3.62318841	-0.173636991	3.65555858	0.0002683031
## 5	permres	0.95937500	0.95605701	-0.003317993	0.24877668	0.8035775682
## 6	nationality	0.85937500	0.90380048	0.044425475	-2.17827834	0.0295861583
## 7	unemployment	0.05312500	0.07838480	0.025259798	-1.49367937	0.1355313860
## 8	statesector	0.06481481	0.03804348	-0.026771337	1.59999765	0.1100114379


```
## 9          ur    0.13750000  0.13539192 -0.002108076  0.09357343  0.9254641782
## 10         comp  0.11562500  0.12232779  0.006702791 -0.31328861  0.7541177195
## 11  opposition  0.16875000  0.15083135 -0.017918646  0.75196435  0.4522250401
```

Appendix III: Differences in partisanship across periods

```
plot.app3.data <- data %>%
  group_by(period123) %>%
  summarize(UR = mean(ur, na.rm = TRUE),
            CPRF = mean(comp, na.rm = TRUE),
            LDPR = mean(ldpr, na.rm = TRUE),
            Opposition = mean(opposition, na.rm = TRUE),
            Undecided = mean(dk, na.rm = TRUE)) %>%
  ungroup() %>%
  pivot_longer(cols = -period123, names_to = "party.name", values_to = "share") %>%
  mutate(party.name = factor(party.name,
                             levels = c("UR", "CPRF", "LDPR", "Opposition", "Undecided")),
         period.name = factor(period123, levels = c(1, 2, 3),
                              labels = c("Pre-Election", "Post-Election", "Post-Protest"))))

ggplot(plot.app3.data, aes(x = party.name, y = share, fill = period.name)) +
  geom_col(position = "dodge") +
  labs(x = NULL, y = "Proportion of Respondents", fill = "Period")
```



Appendix IV: Non-responses

Period 1 vs. Period 2 test (without electday and protestday variables)

```
t3a.trustgovind <- t.test(balance.data1a$trustgovind.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustarmy <- t.test(balance.data1a$trustarmy.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustpolice <- t.test(balance.data1a$trustpolice.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustfsb <- t.test(balance.data1a$trustfsb.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustcourt <- t.test(balance.data1a$trustcourt.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustmungov <- t.test(balance.data1a$trustmungov.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustfedgov <- t.test(balance.data1a$trustfedgov.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustduma <- t.test(balance.data1a$trustduma.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustprocuracy <- t.test(balance.data1a$trustprocuracy.nr ~ balance.data1a$period1min, var.equal = TRUE)
t3a.trustun <- t.test(balance.data1a$trustun.nr ~ balance.data1a$period1min, var.equal = TRUE)

tests.3a <- list(gov.index = t3a.trustgovind, army = t3a.trustarmy, police = t3a.trustpolice, fsb = t3a.trustfsb,
                 court = t3a.trustcourt, mun.gov = t3a.trustmungov, fed.gov = t3a.trustfedgov,
                 дума = t3a.trustduma, procuracy = t3a.trustprocuracy, un = t3a.trustun)

balance.tab3a <- data.frame(var.name = names(tests.3a),
                             mean.period2 = sapply(tests.3a, function(x) x$estimate[1]),
                             mean.period1 = sapply(tests.3a, function(x) x$estimate[2]),
                             diff = sapply(tests.3a, function(x) x$estimate[2] - x$estimate[1]),
                             t.stat = sapply(tests.3a, function(x) x$statistic),
                             p.val = sapply(tests.3a, function(x) x$p.value),
                             row.names = NULL)
```

balance.tab3a

##	var.name	mean.period2	mean.period1	diff	t.stat	p.val
## 1	gov.index	0.818750	0.919028340	0.10027834	-3.4675748	0.0005651464
## 2	army	0.025000	0.008097166	-0.01690283	1.5165676	0.1299351607
## 3	police	0.018750	0.004048583	-0.01470142	1.5725781	0.1163763252
## 4	fsb	0.100000	0.036437247	-0.06356275	2.9141040	0.0037083624
## 5	court	0.053125	0.020242915	-0.03288209	2.0139595	0.0444872880
## 6	mun.gov	0.046875	0.012145749	-0.03472925	2.3459874	0.0193214736
## 7	fed.gov	0.028125	0.016194332	-0.01193067	0.9402111	0.3475110755
## 8	duma	0.046875	0.020242915	-0.02663209	1.7059245	0.0885717168
## 9	procuracy	0.053125	0.016194332	-0.03693067	2.3156883	0.0209324483
## 10	un	0.200000	0.113360324	-0.08663968	2.7886381	0.0054713895

Period 1 vs. Period 3 test (without electday and protestday variables)

```
t3b.trustgovind <- t.test(balance.data1b$trustgovind.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustarmy <- t.test(balance.data1b$trustarmy.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustpolice <- t.test(balance.data1b$trustpolice.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustfsb <- t.test(balance.data1b$trustfsb.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustcourt <- t.test(balance.data1b$trustcourt.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustmungov <- t.test(balance.data1b$trustmungov.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustfedgov <- t.test(balance.data1b$trustfedgov.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustduma <- t.test(balance.data1b$trustduma.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustprocuracy <- t.test(balance.data1b$trustprocuracy.nr ~ balance.data1b$period1min, var.equal = TRUE)
t3b.trustun <- t.test(balance.data1b$trustun.nr ~ balance.data1b$period1min, var.equal = TRUE)

tests.3b <- list(gov.index = t3b.trustgovind, army = t3b.trustarmy, police = t3b.trustpolice, fsb = t3b.trustfsb,
                 court = t3b.trustcourt, mun.gov = t3b.trustmungov, fed.gov = t3b.trustfedgov,
                 дума = t3b.trustduma, procuracy = t3b.trustprocuracy, un = t3b.trustun)
```

```
balance.tab3b <- data.frame(var.name = names(tests.3b),
                           mean.period3 = sapply(tests.3b, function(x) x$estimate[1]),
                           mean.period1 = sapply(tests.3b, function(x) x$estimate[2]),
                           diff = sapply(tests.3b,
                                           function(x) x$estimate[2] - x$estimate[1]),
                           t.stat = sapply(tests.3b, function(x) x$statistic),
                           p.val = sapply(tests.3b, function(x) x$p.value),
                           row.names = NULL)
```

```
balance.tab3b
```

##	var.name	mean.period3	mean.period1	diff	t.stat	p.val
## 1	gov.index	0.78826764	0.919028340	0.13076070	-4.790580	1.848623e-06
## 2	army	0.02566453	0.008097166	-0.01756736	1.684484	9.232178e-02
## 3	police	0.01466544	0.004048583	-0.01061686	1.345173	1.787975e-01
## 4	fsb	0.13290559	0.036437247	-0.09646834	4.316383	1.703029e-05
## 5	court	0.05591201	0.020242915	-0.03566909	2.340611	1.939869e-02
## 6	mun.gov	0.02841430	0.012145749	-0.01626855	1.467183	1.425617e-01
## 7	fed.gov	0.02749771	0.016194332	-0.01130338	1.018973	3.084002e-01
## 8	duma	0.04124656	0.020242915	-0.02100365	1.571872	1.162171e-01
## 9	procuracy	0.06966086	0.016194332	-0.05346653	3.210179	1.358078e-03
## 10	un	0.23098075	0.113360324	-0.11762043	4.126436	3.912131e-05

Period 2 vs. Period 3 test (without electday and protestday variables)

```
t3c.trustgovind <- t.test(balance.data1c$trustgovind.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustarmy <- t.test(balance.data1c$trustarmy.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustpolice <- t.test(balance.data1c$trustpolice.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustfsb <- t.test(balance.data1c$trustfsb.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustcourt <- t.test(balance.data1c$trustcourt.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustmungov <- t.test(balance.data1c$trustmungov.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustfedgov <- t.test(balance.data1c$trustfedgov.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustduma <- t.test(balance.data1c$trustduma.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustprocuracy <- t.test(balance.data1c$trustprocuracy.nr ~ balance.data1c$period3, var.equal = TRUE)
t3c.trustun <- t.test(balance.data1c$trustun.nr ~ balance.data1c$period3, var.equal = TRUE)
```

```
tests.3c <- list(gov.index = t3c.trustgovind, army = t3c.trustarmy, police = t3c.trustpolice, fsb = t3c.trustfsb, court = t3c.trustcourt, mun.gov = t3c.trustmungov, fed.gov = t3c.trustfedgov, duma = t3c.trustduma, procuracy = t3c.trustprocuracy, un = t3c.trustun)
```

```
balance.tab3c <- data.frame(var.name = names(tests.3c),
                           mean.period2 = sapply(tests.3c, function(x) x$estimate[1]),
                           mean.period3 = sapply(tests.3c, function(x) x$estimate[2]),
                           diff = sapply(tests.3c,
                                           function(x) x$estimate[2] - x$estimate[1]),
                           t.stat = sapply(tests.3c, function(x) x$statistic),
                           p.val = sapply(tests.3c, function(x) x$p.value),
                           row.names = NULL)
```

```
balance.tab3c
```

##	var.name	mean.period2	mean.period3	diff	t.stat	p.val
## 1	gov.index	0.818750	0.77553444	-4.321556e-02	1.608885553	0.10791347
## 2	army	0.025000	0.02494062	-5.938242e-05	0.005791672	0.99537994
## 3	police	0.018750	0.01306413	-5.685867e-03	0.720653861	0.47126770
## 4	fsb	0.100000	0.15201900	5.201900e-02	-2.302067133	0.02150783
## 5	court	0.053125	0.05463183	1.506829e-03	-0.101241085	0.91937658

## 6	mun.gov	0.046875	0.02019002	-2.668498e-02	2.487527497	0.01300309
## 7	fed.gov	0.028125	0.02731591	-8.090855e-04	0.075221381	0.94005154
## 8	duma	0.046875	0.03800475	-8.870249e-03	0.685168456	0.49337451
## 9	procuracy	0.053125	0.07363420	2.050920e-02	-1.240420409	0.21507077
## 10	un	0.200000	0.23752969	3.752969e-02	-1.363775149	0.17290292

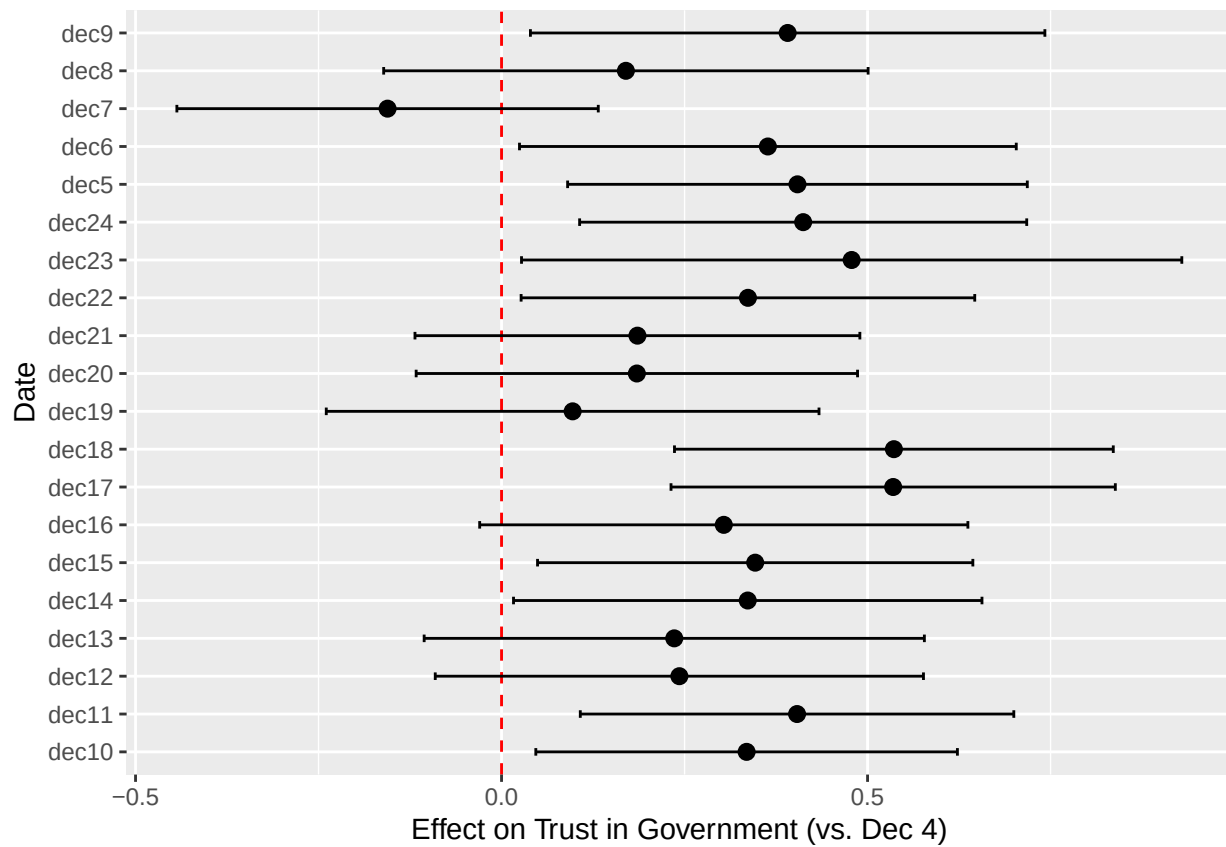
Appendix V: Daily responses

```
data <- data %>%
  mutate(time.trend = case_when(MONTH == 11 ~ day - 24,
                                MONTH == 12 ~ day + 6),
         date = case_when(MONTH == 11 ~ paste0("nov", day),
                          MONTH == 12 ~ paste0("dec", day)),
         date = factor(date, levels = c(paste0("nov", 25:30), paste0("dec", 1:24))),
         date = relevel(date, ref = "dec4"))

trend.reg <- lm(trustgovind ~ date + education + wealth, data = data)
trend.coefs.robust <- coeftest(trend.reg, vcov = vcovHC(trend.reg, type = "HC2"))

dates.to.keep <- paste0("datedec", 5:24)
trend.plot.data <- data.frame(day = paste0("dec", 5:24),
                              beta = trend.coefs.robust[dates.to.keep, "Estimate"],
                              se = trend.coefs.robust[dates.to.keep, "Std. Error"]) %>%
  mutate(ci.lower = beta - 1.96 * se,
         ci.upper = beta + 1.96 * se)

ggplot(trend.plot.data, aes(x = beta, y = day)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  geom_point(size = 2.5) +
  geom_errorbar(aes(xmin = ci.lower, xmax = ci.upper), height = 0.2) +
  labs(x = "Effect on Trust in Government (vs. Dec 4)", y = "Date")
```



Online appendices

Data manipulation

Online Appendix 1

Online Appendix 2

Online Appendix 3

Online Appendix 4